VERITRUST

IMPLEMENTATION HANDBOOK | 11 AUGUST 2026

Deepfake Detection Architecture Blueprint

A complete, evidence-first implementation plan for models, media safety, training, calibration, deployment, scaling, observability and release.

- Two-task model strategy: facial manipulation and fully synthetic imagery
- Hugging Face candidate portfolio and model registry worksheet
- Interactive table of contents, links, fields and completion checklist
- Repository-grounded implementation sequence and acceptance tests

Status: implementation-ready design; model candidates remain benchmark-gated

Scope: image detection only; audio/video/C2PA are intentionally deferred

How to use this handbook

This is a build specification, not a performance claim

No probabilistic detector can be guaranteed 100% correct. Every model named here is a candidate until VeriTrust validates its exact revision on a frozen, legally usable, production-representative benchmark. A failed, unsupported, low-quality or out-of-distribution result must never be converted into safe evidence.

The document is interactive. Use the clickable table of contents and PDF outline to navigate. External references are clickable. The final operator worksheet contains fillable fields, and the release checklist contains interactive checkboxes.

Decision that must be made first

VeriTrust currently uses the words deepfake, synthetic media and fake image as if they were one task. They are not. The implementation must maintain two separately benchmarked tracks:

Track	Question answered	Input	Primary evidence
A - Facial manipulation	Was a real face or identity locally manipulated, swapped or reenacted?	One or more aligned face child artifacts	FACE_MANIPULATION_SCORE
B - Fully synthetic image	Was the complete image generated by an AI image generator?	Deterministically normalized full frame	SYNTHETIC_IMAGE_SCORE

The existing Pixel and Prism names may remain product aliases, but each alias must resolve through the Model Registry to one exact task, revision, preprocessing pipeline, label map, calibration package and benchmark version.

Recommended reading order

- Executives/product: Sections 1, 3, 5, 16 and 20.
- Backend engineers: Sections 4, 7-13 and 16-18.
- ML engineers: Sections 5-6 and 14-15.
- Security/operations: Sections 7, 17-20.

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1. Executive implementation decision

The existing application has credible foundations: protected uploads, basic byte and MIME checks, hashing, private object storage, asynchronous Gateway jobs, retention handling, rate limits and some model metadata. It is not yet the high-assurance deepfake system described in the VeriTrust Four Core Detection Architecture Blueprint.

Release blockers found in the repository

Priority	Finding	Required correction
P0	Unknown provider labels are guessed from text or LABEL_1-like values.	Exact per-revision output label map; reject everything else.
P0	Registry metadata is selected after provider inference.	Resolve immutable model version before preprocessing or inference.
P0	A failed selected model silently switches to the other model.	Emit FAILED for that model; run a second detector only as independent evidence.
P0	Compressed bytes are bounded, decoded pixels and frames are not.	Header limits plus bounded decode in an isolated media worker.
P0	Raw score is used as probability, confidence, risk and verdict.	Keep raw score, calibrated probability and decision confidence separate.
P1	Face crop differs between web, developer API and Gateway.	One versioned server-side preprocessing pipeline.
P1	No quality/OOD/abstention pipeline exists.	Quality evidence and first-class UNCERTAIN/UNSUPPORTED/FAILED states.
P1	No deepfake behavioral, benchmark or robustness tests exist.	Provider fixtures, media security corpus, golden preprocessing and benchmark harness.

Non-negotiable scientific rules

- A specialist model produces typed evidence. It never produces ALLOW or BLOCK.
- LIKELY_REAL means evidence leans real within a qualified benchmark scope. It never certifies authenticity.
- Missing evidence, provider failure or an unsupported input is not negative evidence.
- No threshold is promoted until held-out calibration and release-delta tests pass.
- Model disagreement is stored, displayed and used to reduce sufficiency.
- Same artifact + preprocessing version + model revision must be reproducible.

2. Scope, terminology and trust boundary

In scope

- JPG, PNG and WebP still images after explicit capability validation.
- Full-frame AI-generated-image detection.
- Face discovery, alignment and image-level face manipulation detection.
- Quality measurement, limited OOD handling, calibration, disagreement and abstention.
- Hugging Face Hub-hosted model artifacts, dedicated endpoints or self-hosted inference.
- Direct web API, developer API and Gateway integration through one canonical engine.

Explicitly deferred

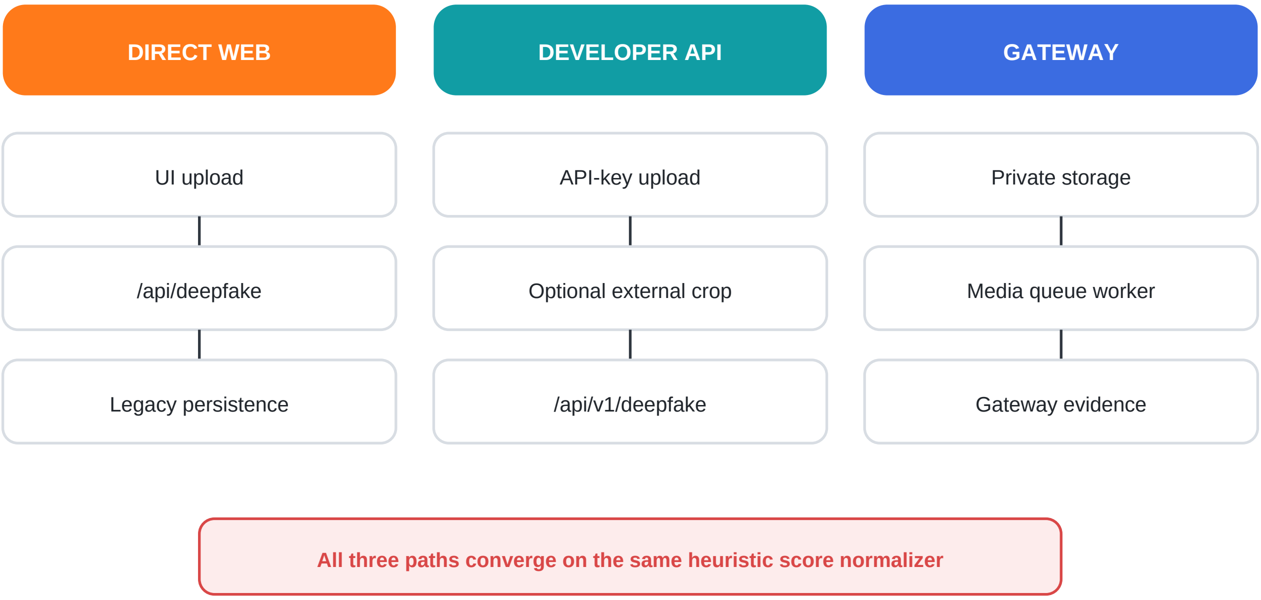
- Video temporal models, audio deepfake detection, C2PA/provenance and source attribution.
- Generator-family attribution, localization masks as forensic proof and biometric identity recognition.

• Automated punitive action based only on a deepfake model.

Vocabulary

Term	Definition
Raw score	A value emitted by a declared model head. It is not automatically a probability.
Calibrated probability	A versioned transformation evaluated on held-out data for a declared scope.
Quality	Measurable image conditions that affect model validity or evidence sufficiency.
OOD	Input outside the qualified content/generator/transformation regimes. Heuristics alone cannot prove OOD.
Evidence sufficiency	Whether required evidence classes and quality are present for a scoped verdict.
Decision confidence	Gateway-level confidence in a policy decision; never copied from one model score.

3. Repository-grounded current architecture



The direct web route and developer API call the same detection service. The Gateway downloads private media in a worker and then calls an adapter that reaches the same detection service. This reuse is valuable, but the shared core currently normalizes unknown model labels heuristically and treats the second detector as a fallback.

Relevant files

Area	File	Current responsibility
Model aliases/provider	lib/veritrust-api.js	Pixel/Prism names, Hugging Face requests and multipart parser.
Detection core	lib/detection-service.js	Heuristic label normalization, model fallback and legacy result.
Input validation	lib/validators.js	4 MB, declared MIME and basic magic-byte checks.
Web route	lib/routes/detection/deepfake.js	Authenticated saved scan and SHA-256.
Developer API	lib/routes/v1/deepfake.js	API-key scan and optional external crop service.
Gateway adapter	lib/models/deepfake-image.js	Maps fake score to safe/suspicious/manipulated.
Media worker	worker/handlers/media.js	Downloads, validates, hashes, runs adapter and finalizes.
Registry metadata	lib/gateway/persistence.js	Selects gateway_model_versions while recording evidence.
UI	assets/js/pages/detection.js	Crop, submit and probability-style result rendering.

What should be preserved

- Existing authentication, API-key scope, rate limit and entitlement boundaries.
- Private object storage, retention jobs, queue leases, idempotency and cancellation foundations.
- Provider request size/time limits and response redaction.
- The public model-performance page's honest statement that no benchmark is currently published.

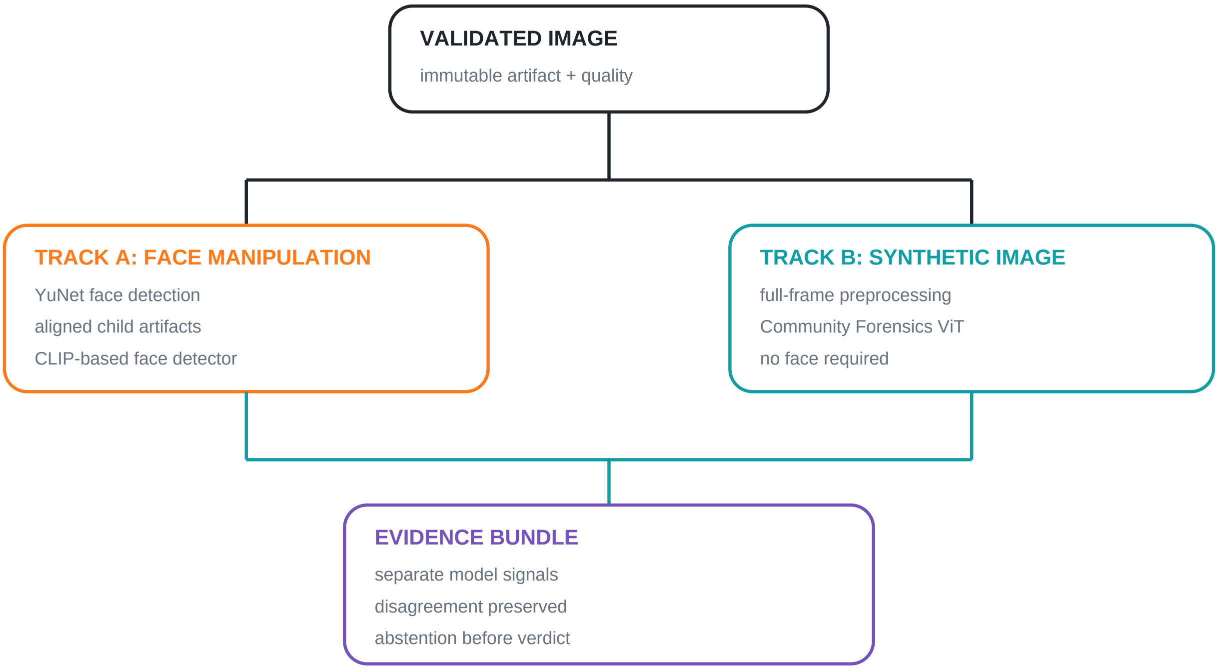
4. Target deepfake architecture

End-to-end invariant

Resolve before execution

The registry is read before preprocessing. The resolved immutable ModelVersion object drives the exact task, revision, input size, crop policy, normalization, label map, calibration and benchmark metadata. The same object is persisted with the model run.

Two-track model topology



Track A and Track B may both run for a face image, but their observations remain distinct. Their values are not averaged. A future Gateway correlation rule can reason over relationships and disagreement.

5. Hugging Face model portfolio and selection

> **Candidate does not mean approved**
>
> The following models are a benchmark shortlist as of 11 August 2026. Production use requires license review, supply-chain scanning, exact commit pinning, local reproduction, task validation, calibration and VeriTrust benchmark approval. The hidden Pixel/Prism IDs cannot be recovered from the repository because they are intentionally server configuration.

Recommended portfolio

Role / proposed key	Candidate	Why shortlist it	Deployment status
Face detector face_detector_yunet	opencv/opencv_zoo - YuNet ONNX	Small, fast face boxes + landmarks; MIT in model directory. Preprocessing only, never deepfake evidence.	Benchmark and pin exact ONNX SHA-256.
Face manipulation primary face_clip_l14_v1	yermandy/deepfake-detection	CLIP ViT-L/14 method aimed at cross-benchmark, partial facial manipulation generalization.	Self-hosted candidate. Convert/scan weights; not available from HF Inference Providers.
Synthetic primary synthetic_commfor_384	OwensLab/commfor-model-384	Official 21.8M-parameter Safetensors model associated with Community Forensics; 384-pixel track.	Recommended first synthetic benchmark candidate; self-hosted.
Synthetic latency/shadow synthetic_commfor_224	OwensLab/commfor-model-224	Official 21.7M-parameter lower-input companion for latency/cost comparison.	Shadow candidate; promote only if slice gates pass.
Independent forensic secondary forensic_frequency_v1	Train and upload a private VeriTrust model	SRM/DCT/noise-residual branch provides a different feature family from semantic ViT/CLIP models.	Train only if it materially improves selective risk and disagreement handling.
Quality/OOD head media_quality_ood_v1	Deterministic metrics + trained embedding OOD head	No generic off-the-shelf OOD model can be trusted for VeriTrust's production distribution.	Build after collecting qualified train/calibration/OOD splits.

How Pixel and Prism should work

Keep Pixel and Prism only as customer-facing aliases. Do not store their display names as contract keys. A possible mapping is shown below; the final mapping depends on the exact models already configured in deployment.

Display alias	Registry deployment	Allowed task	Behavior
VeriTrust Pixel	synthetic_commfor_384@	fully_synthetic_image	Full-frame evidence only.
VeriTrust Prism	face_clip_l14_v1@	face_manipulation_image	Requires one or more qualified face artifacts.

If the currently deployed Pixel/Prism models have different scientific tasks, preserve their real task declarations and change the UI rather than forcing the aliases into this proposed mapping.

Model parameter cards

A. OpenCV YuNet face detector

Parameter	Starting contract
Purpose	Find face boxes and five landmarks; create child artifacts. It does not classify authenticity.
Artifact	opencv/opencv_zoo/models/face_detection_yunet/face_detection_yunet_2026may.onnx or a vetted 2023mar revision.
Format/runtime	ONNX; OpenCV DNN or ONNX Runtime CPU worker.
Input	Deterministically oriented RGB image; dynamic or registry-declared detector input size.
Detection threshold	Start at 0.8 for evaluation; select from recall/precision testing, not preference.
NMS threshold	Start at 0.3; freeze in preprocessing_version.
Top faces	Maximum 10; each qualified face becomes a child artifact.
Minimum face	Start at 96 x 96 source pixels; below threshold becomes LOW_FACE_RESOLUTION.
License/supply chain	Pin commit and ONNX SHA-256; archive license and upstream source.

B. Face manipulation primary - CLIP ViT-L/14 candidate

Parameter	Starting contract
Hub candidate	yermandy/deepfake-detection, pinned to a verified commit after weight scanning/conversion.
Backbone	openai/clip-vit-large-patch14; 224 x 224 CLIP image input unless the pinned project declares otherwise.
Task	Binary facial manipulation evidence on aligned/cropped faces, not generic AI-image detection.
Preprocessing	Use exact project/pinned processor. Freeze resize, crop, RGB conversion, mean/std and interpolation.
Output map	Must be extracted from exact code/checkpoint and proven by golden fixtures. No generic LABEL_0/LABEL_1 assumption.
Runtime	Dedicated GPU container; do not route through generic HF serverless inference because the model is not provider-deployed.
Weight format	Prefer Safetensors. Treat pickle checkpoints and TorchScript as untrusted artifacts; scan and convert in isolation.
Calibration	Separate temperature/isotonic transform qualified on untouched face-manipulation calibration data.

C. Synthetic-image primary - Community Forensics 384

Parameter	Starting contract
Hub candidate	OwensLab/commfor-model-384, exact commit and Safetensors SHA-256 pinned.
Architecture	ViT-small family; approximately 21.8M parameters; 384-pixel model track.
Task	Fully AI-generated image vs real image. Do not use as proof of local face manipulation.
Input	Full frame; exact official preprocessor/config at pinned revision is authoritative.
Labels	Freeze exact head semantics in registry and test with reference fixtures before any production call.
Runtime	Dedicated CPU/GPU benchmark first; GPU worker recommended for predictable tail latency.
Training lineage	Community Forensics model family; candidate card lists Safetensors and MIT model license.
Calibration	Per-version calibration on VeriTrust production-like data; do not reuse model-card thresholds.

D. Independent forensic secondary

Do not choose a second ViT merely because it has a different repository name. Independence should be measured by error correlation. If a secondary detector is needed, train a compact branch over SRM/high-pass residuals, DCT statistics or multi-scale noise residuals, then prove it catches different failures. Upload the resulting Safetensors/ONNX artifact to a private Hugging Face repository and register its exact commit.

6. Authoritative Model Registry

Required registry fields

Field	Requirement
model_key	Stable internal key; never a display name.
task	face_manipulation_image, fully_synthetic_image, face_detection or quality_ood.
provider	hf_dedicated, self_hosted_onnx, self_hosted_torch or approved alternative.
repository + revision	Exact Hugging Face repository and immutable commit SHA; never main.
artifact_sha256	Digest of every downloaded weight/config/processor artifact.
input_contract	MIME, dimensions, frames, colorspace, crop policy and processor details.
output_label_map	Exact raw provider label/index to VeriTrust observation semantics.
preprocessing_version	Immutable code/config version and deterministic output contract.
calibration_version	Transform, fitting manifest and validated thresholds; null until qualified.
benchmark_version	Frozen evaluation manifest and report checksum; null until qualified.
deployment_status	disabled, shadow, canary, production or rolled_back.
rollout	Deterministic cohort, start/end, target share and rollback predecessor.
license + approval	License text checksum, commercial-use review and approver.

Registry resolution flow

Request -> capability check -> task routing -> registry resolution -> rollout cohort -> exact preprocessing -> exact inference adapter -> evidence persistence. If any registry field needed for execution is missing, the result is UNSUPPORTED or FAILED before provider inference.

Recommended implementation: a shared model_versions table plus immutable calibration_versions and benchmark_versions. If the deployed gateway_model_versions table is retained, create a compatibility view or migration; do not maintain two independent registries.

7. Media ingestion and bounded validation

Initial capability defaults

Control	Synchronous path	Asynchronous object-storage path	Failure state
Request bytes	4 MiB maximum including multipart safety margin	20 MiB initial maximum	IMAGE_TOO_LARGE
Formats	JPEG, PNG, static WebP	JPEG, PNG, static WebP	UNSUPPORTED_MEDIA_TYPE
Minimum dimensions	64 x 64 full frame	64 x 64 full frame	RESOLUTION_TOO_LOW
Maximum side	8,192 pixels	8,192 pixels	DIMENSION_LIMIT_EXCEEDED
Maximum pixels	40 megapixels	40 megapixels	PIXEL_LIMIT_EXCEEDED
Frames/pages	Exactly 1	Exactly 1	ANIMATED_MEDIA_UNSUPPORTED
Channels	1-4 decoded; normalized to RGB	1-4 decoded; normalized to RGB	CHANNEL_LIMIT_EXCEEDED
Decoded budget	160 MiB maximum	160 MiB maximum	DECODE_BUDGET_EXCEEDED
Metadata budget	256 KiB	256 KiB	METADATA_LIMIT_EXCEEDED
Validation deadline	3 seconds	3 seconds	MEDIA_VALIDATION_TIMEOUT

These are safe starting defaults, not eternal constants. Publish them through /api/system/capabilities and change them only through versioned configuration plus security/load regression.

Validation sequence

- Authenticate and authorize before accepting or registering content.
- Assign artifact_id; stream SHA-256 while receiving or downloading bytes.
- Validate declared MIME, extension when relevant and independent file signature.
- Read bounded metadata with safety features enabled; reject animation/multipage content.
- Enforce width, height, pixel, channel and metadata budgets before decode.
- Perform one bounded decode in an isolated media worker; capture warnings as evidence.
- Never log raw bytes, EXIF, faces, original filename or signed object URLs.
- Store bytes privately under generated paths and governed retention.

8. Deterministic preprocessing specification

Shared normalization

Step	Versioned behavior
Orientation	Apply EXIF orientation once; record original orientation and remove it from derived output.
Colorspace	Convert to sRGB using a pinned image library/runtime version.
Alpha	Flatten against a declared background color; do not silently discard transparency.
Bit depth	Convert through a declared path to 8-bit RGB for models that require it.
Metadata	Strip nonessential metadata from derived model input; retain only governed evidence metadata.
Resize	Model-specific size, aspect policy, interpolation and crop; all registry-controlled.
Output hash	SHA-256 of canonical derived bytes plus a preprocessing manifest checksum.

Track A - face manipulation

- Run the pinned face detector on the normalized full frame.
- Create a parent-child artifact edge for every qualified face, up to the configured cap.
- Use five landmarks for alignment when the model contract requires it.
- Apply a fixed margin (start 0.15) and deterministic padding/resize policy.
- Never silently choose face index zero as the only evidence when multiple faces exist.
- If there is no face, emit NO_QUALIFIED_FACE; do not fall back to a generic face model on the full image.

Track B - fully synthetic image

- Use the whole normalized image; face detection is not required.
- Follow the pinned official processor exactly, including resize/crop and normalization.
- Record transformations that may remove forensic traces, including screenshot/re-encode and aggressive resize.

Golden reproducibility test

Maintain a small legal fixture set. For every preprocessing version, CI verifies exact output dimensions, metadata manifest and SHA-256 on the supported production runtime. A hash change requires a new preprocessing version and benchmark delta.

9. Quality, OOD and evidence sufficiency

Quality evidence families

Family	Measurements	Example reason codes
Decode	decoder warnings, truncated data, color conversion, frame count	DECODE_WARNING, TRUNCATED_IMAGE
Resolution	full-frame and face dimensions, resize ratio	LOW_RESOLUTION, LOW_FACE_RESOLUTION
Sharpness	variance of Laplacian or qualified equivalent by content slice	EXCESSIVE_BLUR
Compression	format, estimated JPEG quality, blockiness and repeated transcoding indicators	HEAVY_COMPRESSION, REENCODED_MEDIA
Exposure	luminance distribution, clipped dark/light fraction	UNDEREXPOSED, OVEREXPOSED
Face geometry	landmark confidence, pose, occlusion, face-area ratio	FACE_POSE_OOD, FACE_OCCLUDED
Content regime	embedding distance/energy against qualified in-distribution reference	CONTENT_OOD, GENERATOR_REGIME_UNKNOWN

Important OOD limitation

Do not market a heuristic as OOD detection

Blur, low resolution and compression are quality signals. A real OOD system requires a declared reference distribution and held-out OOD evaluation. Start with explicit quality abstention. Add an embedding-distance, density or energy-based OOD head only after it improves selective risk on frozen OOD datasets.

Sufficiency rule examples

- LIKELY_REAL requires validated media, acceptable quality, in-scope task, qualified calibration and no strong disagreement.
- LIKELY_SYNTHETIC requires threshold-qualified model evidence and sufficient quality; the Gateway still owns action.
- Low quality, OOD, missing calibration or conflicting models produces UNCERTAIN.
- Unsupported task or absence of a qualified face for Track A produces UNSUPPORTED.
- Decode/provider/runtime failure produces FAILED with null semantic score.

10. Inference adapter contract

Adapter input

artifact_id, derived_artifact_id, resolved ModelVersion, preprocessing manifest/hash, deadline, cancellation signal, tenant-safe trace context and request idempotency key.

Adapter output

provider status, raw provider schema version, exact labels/scores, normalized observation, latency, attempt number, error class and redacted response reference. The adapter does not calculate final risk or action.

Hugging Face integration rules

- Use repository@commit, not repository@main.
- Validate config, processor and weight SHA-256 at image build or controlled model acquisition.
- Prefer Safetensors/ONNX. Treat pickle-based files as untrusted and convert only inside an isolated build process.

- Use a dedicated Hugging Face endpoint or self-hosted container for models not deployed by Inference Providers.
- Schema-validate every label and finite score. Reject unknown labels, missing expected classes and duplicates.
- Retry only classified transient errors such as 429/selected 5xx, with bounded exponential backoff and jitter.
- Use provider bulkheads, a circuit breaker and per-model concurrency budgets.
- Never retry unsupported output, invalid input or deterministic preprocessing failures.

Error taxonomy

Class	Examples	Evidence behavior
UNSUPPORTED	task mismatch, unknown label, animation, no qualified face	UNSUPPORTED; null semantic probability
FAILED	timeout, provider outage, invalid JSON, weight load failure	FAILED; preserve attempt and provider version
UNCERTAIN	low quality, OOD, disagreement, missing calibration	Completed evidence with abstention
RETRYABLE	429, transient 502/503, lease-safe worker interruption	Retry under job budget; never return safe while pending

11. Normalization, calibration and thresholds

Data fields that must remain separate

Field	Meaning	May be null?
raw_score	Exact declared model output after label mapping.	No for successful model output
normalized_score	Deterministic score orientation, for example higher = more synthetic.	Yes
calibrated_probability	Versioned held-out calibration output for a qualified scope.	Yes until validated
quality_score	Versioned aggregation of declared quality measurements.	Yes
evidence_sufficiency	Whether required evidence/quality classes are available.	No
decision_confidence	Gateway policy/correlation confidence.	Always absent from specialist evidence

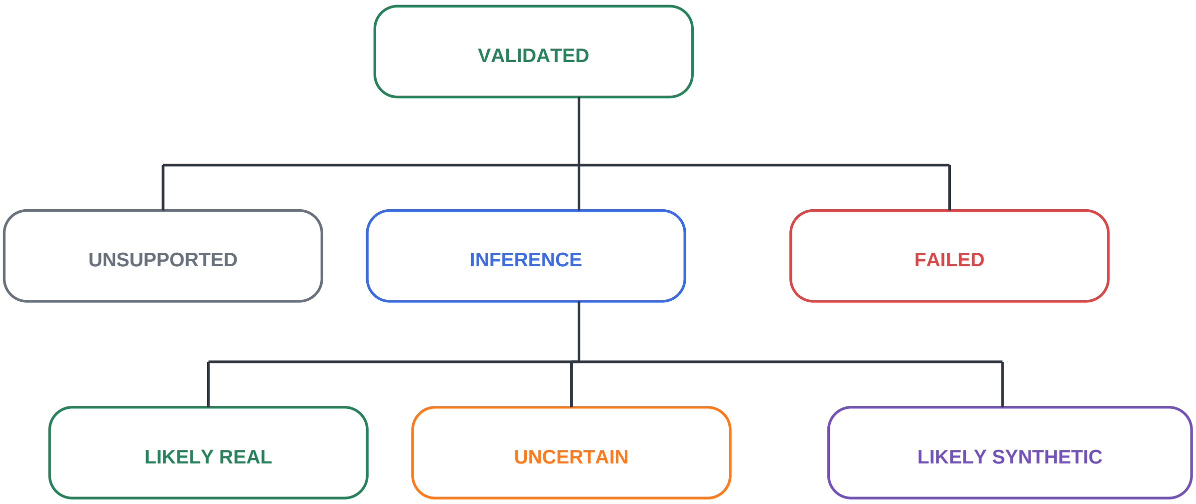
Calibration workflow

- Hold out calibration data by source identity, video, generator family and capture pipeline.
- Compare temperature scaling, Platt/logistic scaling and isotonic regression.
- Select using Brier score, expected calibration error and reliability plots, not accuracy alone.
- Report bootstrap confidence intervals and slice calibration.
- Freeze transform coefficients, fitting manifest/hash and software version.
- Use business costs to choose review/synthetic evidence thresholds; never use arbitrary 0.45/0.75 values.

Before calibration

The API may return raw_score and normalized_score, but calibrated_probability must be null and the evidence verdict should normally remain UNCERTAIN unless a non-probabilistic benchmark-qualified rule explicitly supports another evidence state.

12. Disagreement and abstention state machine



Disagreement policy

- Run a second detector only for policy-selected high-impact requests, shadow evaluation or qualified ensemble routing.
- Persist one evidence object per model. Never overwrite the primary or average scores into false certainty.
- Measure model error correlation before calling a detector independent.
- Strong contradictory calibrated evidence produces UNCERTAIN and mandatory review when the media is material.

Evidence verdict definitions

Verdict	Operational meaning
LIKELY_REAL	Qualified evidence leans real within the declared task/benchmark scope; not authenticity proof.
LIKELY_SYNTHETIC	Qualified evidence leans synthetic/manipulated within the declared scope.
UNCERTAIN	Weak, low-quality, OOD, uncalibrated or conflicting evidence.
UNSUPPORTED	Input/model/task is outside the declared capability contract.
FAILED	Processing/provider failed; no semantic verdict is inferred.

13. Canonical data and API contracts

Artifact example

```
{ "artifact_id": "uuid", "parent_artifact_id": null, "type": "image", "source": "web", "sha256": "hex",
"mime_type": "image/jpeg", "byte_size": 1834210, "retention_class": "temporary_private",
"schema_version": "artifact.v1" }
```

Deepfake Evidence example

```
{ "evidence_id": "uuid", "artifact_id": "uuid", "evidence_type": "synthetic_image_model", "source": {
"adapter": "deepfake_evidence_adapter.v1", "provider": "self_hosted_torch", "model_key":
"synthetic_commfor_384", "revision": "40-char-commit-sha" }, "observation": { "verdict": "UNCERTAIN",
"raw_score": 0.82, "calibrated_probability": null }, "quality": { "status": "PASS", "score": 0.91,
"reasons": [] }, "reason_codes": ["CALIBRATION_NOT_QUALIFIED"], "preprocessing_version":
"fullframe.v1", "calibration_version": null, "benchmark_version": null, "schema_version":
"evidence.deepfake.v1" }
```

Recommended API behavior

Endpoint	Behavior
POST /api/v2/deepfake/investigations	Register artifact/requested tasks; return 202 with durable job for heavy media.
GET /api/v2/deepfake/investigations/{id}	Return status, artifacts, model attempts, evidence and progress.
POST /api/v2/deepfake/investigations/{id}/cancel	Idempotent cancellation request.
GET /api/system/capabilities	Authoritative MIME, byte, pixel, task, model and degraded-state truth.
GET /api/models	Admin/operator registry view; public view exposes approved model cards only.

Existing /api/deepfake and /api/v1/deepfake should become compatibility wrappers over the canonical engine. They may project legacy fields during a deprecation window, but internal truth must be the versioned Artifact/Evidence model.

14. Training and fine-tuning plan

> **Training is conditional, benchmarking is mandatory**
>
> Start by reproducing and benchmarking the shortlisted models. Train or fine-tune only if no candidate meets the declared slice, robustness, calibration, latency and license gates. This avoids spending GPU time before the task and data contract are correct.

Dataset strategy

Dataset/source	Use	Restriction/action
VeriTrust consented production-like corpus	Primary calibration, robustness and final holdout	Required for commercial relevance; consent, provenance, retention and no train/test leakage.
Community Forensics / Small	Synthetic-image research training and reproduction	Dataset card states research use and per-generator licenses; legal review required.
Community Forensics Eval	Synthetic-image research comparison	Non-commercial research/education license; not a commercial production qualification set.
FaceForensics++	Face manipulation research training/benchmark	Terms restrict database to non-commercial research/education; do not use for commercial weights without permission.
Celeb-DF v2	Cross-dataset face manipulation evaluation	Review dataset terms, identity/privacy and redistribution before use.
DFDC	Research cross-dataset evaluation	Subject to competition/data terms; consented paid actors, but commercial rights still require review.
DF40	Diverse research evaluation	CC BY-NC 4.0; research-only for commercial product development unless separately licensed.

Leakage-safe split rules

- Group face data by source identity and original video; derivatives never cross splits.
- Group generated images by generator family/checkpoint and source real image.
- Maintain generator-held-out, manipulation-held-out and transformation-held-out test sets.
- Keep calibration data completely separate from training and final test.
- Hash/near-duplicate scan every split with perceptual and embedding similarity.
- Freeze a final production holdout that model developers cannot inspect sample-by-sample.

Training recipe A - fully synthetic image detector

Parameter	Recommended starting search space
Backbone	Reproduce OwensLab commfor-model-384 first; alternate ViT-S/16 384 only with pinned initialization.
Input	384 x 384 RGB; exact resize/crop and normalization recorded.
Objective	Binary cross entropy/logit or 2-class cross entropy; compare generator-balanced sampling.
Optimizer	AdamW.
Learning rates	Head 1e-4 to 5e-4; backbone 1e-6 to 5e-5. Select via validation, not one fixed value.
Weight decay	0.01 to 0.1.
Effective batch	64-256 using gradient accumulation; log actual per-device batch.
Epochs	10-30 maximum with early stopping on generator-macro AP and selective risk.
Schedule	5% warmup then cosine decay.
Precision	BF16 where validated; FP32 metric accumulation.
Augmentation	JPEG/transcode, resize/crop, screenshot, blur/noise, overlays and platform transformations; do not create label artifacts.
Seeds	At least 3; report mean and confidence interval.
Checkpoint	Best macro AP subject to FPR, calibration and slice floors; not lowest training loss.

Training recipe B - face manipulation detector

Parameter	Recommended starting search space
Backbone	Pinned openai/clip-vit-large-patch14 visual encoder or reproduced yermandy method.
Input	224 x 224 aligned face unless pinned method declares otherwise.
Adaptation	Start with frozen backbone + linear/LN tuning; add PEFT only if cross-dataset validation improves.
Learning rate	5e-5 to 1e-4 for head/LN parameters; lower for any unfrozen backbone blocks.
Effective batch	32-128, balanced by real/fake and manipulation family.
Epochs	5-15 with cross-dataset early stopping; short PEFT schedules may be sufficient.
Loss	Cross entropy; compare focal loss only when imbalance remains after sampling.
Face sampling	Cap frames/faces per source video so long videos do not dominate.
Augmentation	Compression, blur, color, pose-preserving crop jitter, overlay and screenshot; preserve identity grouping.
Validation	Macro across datasets/manipulations, demographic/quality slices and face-detection failure rate.

Training recipe C - independent frequency branch

Use a compact CNN/ConvNeXt-tiny-style backbone over fixed SRM/high-pass residual channels or DCT statistics. Train with the same leakage-safe groups. Its promotion metric is not standalone accuracy; it must reduce correlated errors and improve selective risk when combined by an abstention policy.

Reproducibility package

- Container digest, CUDA/cuDNN/PyTorch/Transformers versions and hardware.
- Dataset manifest hashes, license inventory and exact split membership.
- Config, seed, optimizer/scheduler state, checkpoints and code commit.
- Model card, intended use, limitations, prohibited uses and rollback predecessor.
- SBOM and malicious-model scan results; prefer immutable Safetensors/ONNX.

15. Benchmark, robustness and calibration package

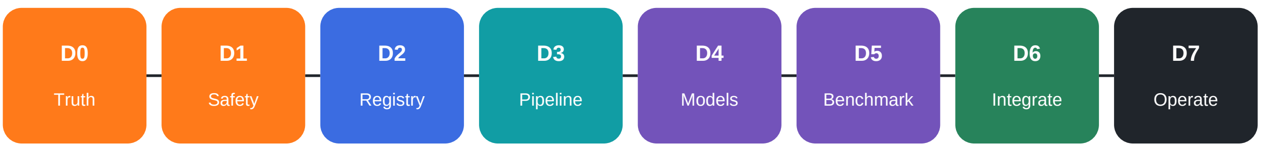
Required metrics

Family	Metrics
Ranking	AUROC, AUPRC, generator/manipulation macro AP.
Operating points	Precision, recall, FPR, FNR, specificity and confusion matrix at every proposed threshold.
Calibration	Brier score, NLL, ECE, reliability curves and confidence intervals.
Abstention	Coverage vs selective risk; error rate among accepted results.
Robustness	Delta under JPEG/transcode, resize/crop, screenshot, blur/noise, overlays and social-platform transforms.
Fairness/slices	Performance by source, camera, face size, pose, lighting, skin-tone/demographic slices where lawful and ethically governed.
Operations	P50/P95/P99 latency, throughput, memory, provider errors, cost/job and fallback/abstention rates.

Promotion gates

- Unknown-label and provider-schema fixtures pass.
- No required slice falls below its declared floor.
- False-positive budget is met on consented production-like real images.
- Robustness delta remains within declared tolerance.
- Calibration and selective-risk targets are met with confidence intervals.
- Latency, queue and memory load tests pass at expected peak plus safety margin.
- Candidate beats or complements the production version and has an instant rollback target.

16. Exact implementation roadmap



Each phase is blocked until its exit gates are green

Phase	Deliverables	Exit gate
D0 - Truth	Freeze tasks, current Pixel/Prism repos/commits/labels/licenses; shared schemas.	Unknown contract fields fail before inference.
D1 - Safety	Restore DB migrations, behavioral tests, bounded media validation and capability registry.	Bomb/malformed corpus rejected before provider.
D2 - Registry	Authoritative registry repository, calibration/benchmark tables, deterministic rollout.	Executed and recorded model revision are identical.
D3 - Pipeline	Deterministic full-frame/face preprocessing, derived artifacts, quality evidence.	Golden output hashes reproduce in CI.
D4 - Models	Strict adapters, candidate reproduction, optional independent secondary and abstention.	All provider schemas and failure states tested.

Phase	Deliverables	Exit gate
D5 - Benchmark	Frozen evaluation, robustness, calibration, thresholds and model cards.	Promotion report approved; otherwise remain shadow.
D6 - Integrate	One core under web, v1 and Gateway; evidence-first UI and compatibility window.	Same artifact/version yields same canonical evidence across entrypoints.
D7 - Operate	Async upload/jobs, SLOs, dashboards, drift, feedback, canary and rollback.	Failure injection/load/rollback drills pass.

First engineering tranche

Ticket	Files/modules	Acceptance
DFK-001 Contracts	lib/contracts/*, runtime validators, fixtures	Artifact/Evidence/ModelVersion schemas reject unknown versions/fields.
DFK-002 Migrations	supabase/migrations/* or governed migration root	Clean bootstrap, upgrade and restore in CI.
DFK-003 Registry	lib/model-registry/*, admin repository	Exact revision resolves before inference; main is prohibited.
DFK-004 Label safety	lib/deepfake/normalizer.js	LABEL_1/unknown/NaN/duplicates produce UNSUPPORTED_MODEL_OUTPUT.
DFK-005 Media safety	lib/media/*, worker validation	Pixels/frames/decode budgets and malformed fixtures pass.
DFK-006 Compatibility	detection-service wrapper	Legacy consumers can project evidence without becoming internal truth.

17. Test architecture

Test layers

Layer	Required cases
Unit	Registry resolution, label mapping, score validation, calibration transform, reason codes and state transitions.
Contract	Known/unknown provider schemas, all Evidence states, API schema versions and compatibility projections.
Media security	Magic mismatch, truncation, pixel bomb, animation, huge metadata, odd channels and decode timeout.
Golden preprocessing	Orientation, alpha, colorspace, resize, face crops, multi-face relationships and output SHA-256.
Integration	Object storage -> queue -> worker -> provider fixture -> immutable evidence -> status API.
Failure injection	Provider 429/5xx/timeout/schema change, registry outage, lease loss, cancellation and DB retry.
Benchmark	Frozen manifests, metrics, calibration, slice/robustness deltas and release report.
E2E	Web and developer API render UNCERTAIN/UNSUPPORTED/FAILED without low-risk substitution.

CI lanes

- Every commit: syntax/lint, unit, contract, security fixtures and small golden corpus.
- Pull request: integration tests with local provider fixtures and clean database migration.
- Scheduled/GPU: robustness suite, candidate benchmark and drift report.
- Release: full benchmark delta, supply-chain scan, load/failure drill and rollback rehearsal.

18. Deployment and horizontal scaling

Recommended production topology

Client -> authenticated API -> signed private object upload -> immutable artifact registration -> durable media queue -> validation/preprocessing worker -> per-model inference pools -> immutable evidence -> polling/webhook -> Gateway decision later.

Service responsibilities

Component	Responsibility	Scaling signal
API/Vercel	Auth, quotas, capabilities, upload registration, job/status APIs.	Request rate; no heavy decode/inference.
Object storage	Private encrypted artifacts and short retention.	Storage operations and retention backlog.
Media CPU workers	Hash, bounded decode, quality and deterministic preprocessing.	Queue depth, CPU and memory.
Face detector pool	YuNet/approved detector and derived artifact creation.	Face-task queue and CPU latency.
GPU inference pools	One pool per compatible model/runtime; batched where validated.	Queue age, GPU utilization and tail latency.
Evidence store	Immutable registry/model runs/evidence and audit references.	Write latency, orphan checks and index health.

Backpressure and reliability

- Tenant/global quotas and maximum in-flight jobs.
- Separate queues or priorities for preprocessing and expensive model execution.
- Per-provider/model semaphore, circuit breaker and bounded retry budget.
- Lease heartbeat, idempotency and cancellation at stage boundaries.
- Preliminary status remains HOLD/PENDING; missing model evidence never becomes safe.
- Autoscale on queue age/depth and measured service time; cap scale to provider/database capacity.

SLOs to establish through load testing

- API registration availability and latency.
- P95 queue age and end-to-end job latency by model/task.
- Maximum provider error and timeout rate.
- Cancellation and retention completion latency.
- Evidence persistence integrity and orphan rate.

19. Observability, drift and feedback

Metrics

Family	Examples
Reliability	accepted/completed/failed/unsupported/uncertain, retries, queue age, cancellation and worker lease loss.
Providers/models	latency, 429/5xx/schema rejection, model revision distribution, circuit state and batch size.
Media quality	resolution, face detection success, blur/compression buckets and quality abstention.
Scores	raw/calibrated distributions by model, preprocessing and quality slice; never raw images.
Decisions later	Gateway action distribution by policy/correlation/model versions.
Feedback	analyst labels, appeals, confirmed FP/FN, time-to-review and feedback coverage.
Integrity	orphan evidence, revision mismatch, missing hashes, retention failures and migration drift.

Drift rules

- Alert on provider output schema or label-set changes immediately.
- Track PSI/KS or declared distribution checks on quality, embedding and score features.

- Separate population drift from model regression and upstream preprocessing changes.
- Use delayed human/verified feedback to estimate production error; never treat model agreement as truth.
- Trigger shadow re-evaluation before retraining; data drift alone does not authorize automatic model updates.

20. Security, privacy, licensing and release governance

Security controls

- Run image decode and third-party model loading in isolated, least-privilege workers.
- Use private storage, generated paths, short retention and verified deletion receipts.
- Never expose service credentials, provider model paths, signed URLs or raw content in logs.
- Scan model artifacts; prefer Safetensors/ONNX; maintain SBOM and pinned dependency lockfiles.
- Separate model build/training credentials from production inference identity.
- Rate-limit upload registration, status polling and API-key usage by tenant.
- Require explicit user authority to process faces; document biometric/privacy considerations.

Licensing gate

A permissive model-code license does not automatically grant commercial rights to training data or pretrained weights. The registry must store model, code, dataset and upstream-weight licenses separately. FaceForensics++, DF40 and Community Forensics evaluation resources contain non-commercial/research restrictions. Obtain permission or build a consented/licensed commercial corpus before training production weights.

Definition of done

- Every output can be reproduced from artifact, preprocessing, model, calibration and benchmark versions.
- No missing provider, parser failure, unknown label or disagreement becomes safe evidence.
- Every promoted model has a signed benchmark delta and rollback predecessor.
- UI and exports explain uncertainty and scope without an unexplained confidence percentage.
- Clean migration, security, resilience, load and rollback gates pass in CI/release rehearsal.

21. Interactive operator worksheet

Fill these fields before DFK-003 is implemented. The exact currently deployed model identifiers are intentionally not present in the repository.

Current Pixel deployment

Hugging Face repository	
Exact commit SHA	
Scientific task	
Raw labels / indices	

Preprocessor / input	
License approval	

Current Prism deployment

Hugging Face repository	
Exact commit SHA	
Scientific task	
Raw labels / indices	
Preprocessor / input	
License approval	

Target operating context

Primary customers/use	
False-positive budget	
False-negative budget	
Peak scans / minute	
Maximum job latency	
Retention requirement	

22. Interactive implementation and release checklist

Truth and contracts

- [ ] Pixel exact repository, commit, task, label map and license recorded
- [ ] Prism exact repository, commit, task, label map and license recorded

- [ ] Artifact/Evidence/ModelVersion schemas and runtime validators merged
- [ ] Clean database bootstrap, upgrade and restore pass
- [ ] Registry selects exact revision before preprocessing/inference
- [ ] Unknown/duplicate/non-finite provider output fails closed
- [ ] Dimension, pixel, frame, metadata and decode budgets enforced
- [ ] Preprocessing golden hashes reproduce in CI
- [ ] All qualified faces become related child artifacts
- [ ] Quality evidence and sufficiency rules implemented
- [ ] OOD claim is benchmarked or omitted
- [ ] Provider adapters preserve attempts, errors and exact versions
- [ ] Second model is independent evidence, not silent fallback
- [ ] LIKELY_REAL/LIKELY_SYNTHETIC/UNCERTAIN/UNSUPPORTED/FAILED implemented
- [ ] Frozen held-out benchmark and robustness report approved
- [ ] Calibration transform and thresholds approved
- [ ] UI/API display failure and uncertainty without low-risk substitution
- [ ] Signed upload, queue, backpressure and cancellation load-tested
- [ ] Provider/model/quality/drift dashboards and alerts live
- [ ] Shadow, deterministic canary and instant rollback rehearsed

23. Sources and primary references

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S19. DF40 official repository and CC BY-NC license statement. https://github.com/YZY-stack/DF40

S20. Community Forensics official training/evaluation code. https://github.com/JeongsooP/Community-Forensics

Final implementation starting point

Begin with DFK-001 through DFK-005: freeze Pixel/Prism semantics, restore migrations, introduce authoritative schemas/registry, remove heuristic labels and add bounded media validation. Do not implement training, calibration or a second detector before these correctness foundations pass.

END OF HANDBOOK